Project Analysis Report: Pizza Sales 2015

Following the pizza sales performance data cleaning, transformation and visualisation, the following analysis was made:

1. Sales Performance & Revenue Analysis

- **Total Revenue:** The shop generated in **$ 817 860.05** in 2015.
- **Monthly Trends:** Revenue is highest in **July** and **May**. Significant dips were identified in **September and October**, suggesting a need for "Back-to-School", "Halloween", or “Spring Festival” promotions. These can be in the form of bundle deals or Buy-One-Get-One-Free (BOGOF) deals.
- **Peak Distribution:** Sales peak on **Fridays**, particularly during the "Lunch Rush" (**12 PM-1 PM**) and "Dinner Rush" (**5 PM-7 PM**).
- **Average Order Value (AOV):** The average customer spends **$38.31** per transaction.

2. Product Performance & Menu Optimization

- **Top Sellers:** The **Classic Deluxe Pizza** is the most popular by quantity, while the **Thai Chicken Pizza** drives the highest revenue due to its premium price point.
- **Bottom Sellers:** The **Brie Carre Pizza** has the lowest sales volume, management should consider a "Limited Time Offer" strategy or removing it to reduce inventory waste.
- **Size Preference: Large (L)** pizzas contribute the highest proportion of revenue, accounting for nearly 46% of total sales.

3. Pricing & Discount Strategy

- **Price Sensitivity:** Analysis shows that while <u>premium pizzas (Chicken/Supreme)</u> sell well, there is a clear drop in volume when unit prices exceed **$20**.
- **High-Value Orders:** Orders over $50 (High-Value) contribute to 48.4%of total revenue despite being fewer in number.
- **Recommendation:** Implement a **Loyalty Program** specifically targeting these high-value customers to maintain retention during slow months (Feb/Sept).

4. Ingredient & Inventory Optimization

- **Key Ingredients: Garlic, Tomatoes, and Red Onions** are the most frequently used items.

- **Supply Chain:** To ensure freshness and prevent stockouts during Friday peaks, inventory for these "Core Three" ingredients should be replenished no later than Thursday morning.
- **Profitability:** Using an estimated 30% food cost, the **Classic category** maintains the healthiest margins, while **Veggie pizzas** offer the lowest ingredient cost-to-revenue ratio.

5. Operational Efficiency

- **Staffing:** Based on order density, the kitchen requires maximum staffing on **Fridays from 4 PM onwards**.
- **Data Gaps:** Currently, "Order Processing Time" and "Service Channel" (Dine-in vs. Delivery) are not captured.
- **Future Goal:** To improve operations, I recommend integrating a timestamp for "Order Ready" to calculate the average prep time per pizza type.

Summary

The data reveals a robust business with strong weekend performance. By shifting focus toward boosting mid-week sales through targeted promotions in September and October, and prioritizing inventory for high-volume ingredients like Garlic, the business can realistically target a 5%-10% increase in annual revenue for 2016."

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